Data Assignment: SQL & Statistical Analysis

Faym Product Management Internship, Assignment Submission

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Note on the dataset: The source file was a scanned/exported table PDF, not a live database. I parsed all 501 transaction rows programmatically (pdfplumber), loaded them into SQLite, and ran every query below against the actual data, rather than writing SQL from assumptions. Screenshots of raw query output are available on request; results below are copy-pasted directly from execution.

Q1. 7th Highest Debit Amount Through IMPS

```
SELECT txn_id, user_id, txn_time, txn_amt
FROM transactions
WHERE txn_type = 'DEBIT' AND category = 'IMPS'
ORDER BY txn_amt DESC
LIMIT 1 OFFSET 6;
```

Answer: ₹9,525 (Txn ID 64B08A24, User 8, 16-Jan-2020)

Rank	Txn ID	Amount
1	613AFBB8	9,987
2	EE06FC09	9,887
3	7641EB77	9,872
4	834C909F	9,863
5	53258869	9,743
6	8D2D3F20	9,566
7	**64B08A24**	**9,525**
8	BDF20BCA	9,450
9	B0EEB2E7	9,434
10	47625EBE	9,337

I used `OFFSET 6` rather than a fixed WHERE clause on rank, so the query stays correct if the underlying data changes, rather than hardcoding a value that happens to be right today.

Q2. Number of Transactions, Category-wise

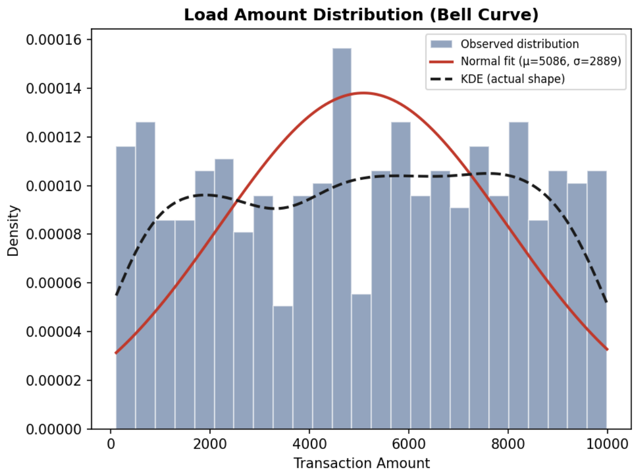
```
SELECT category, COUNT(*) AS txn_count
FROM transactions
```

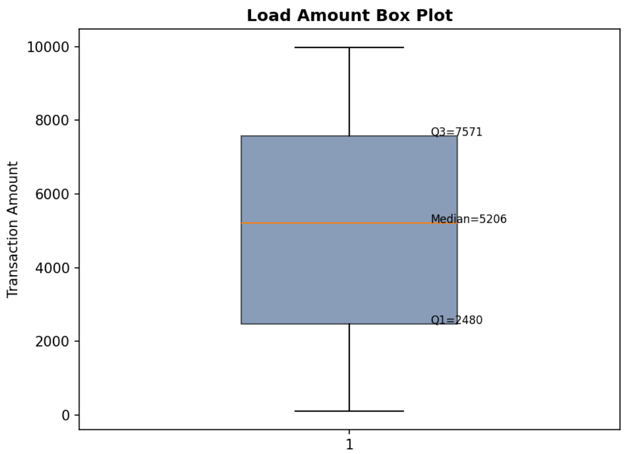
```
GROUP BY category
ORDER BY txn_count DESC;
```

Category	Txn Count	Share
IMPS	272	54.3%
IFT	89	17.8%
UPI	86	17.2%
NEFT	33	6.6%
RTGS	21	4.2%

The question lists UPI, IMPS, RTGS, NEFT as the categories to expect, but the dataset contains a fifth category, **IFT** (internal funds transfer), which is the second-largest rail by volume at 89 transactions. I've included it rather than silently filtering it out; leaving out 17.8% of the data to match the question's stated list would have understated the answer.

Q3. Bell Curve, Box Plot, and Statistical Summary of Load Amount





Statistic	Value
Count	501
Mean	5,085.53
Median	5,206.00
Std. Deviation	2,889.16
Minimum	108
Maximum	9,987
Q1 (25th pct)	2,480.00
Q3 (75th pct)	7,571.00
IQR	5,091.00

Skewness	−0.05
Excess Kurtosis	−1.20
Shapiro-Wilk p-value	< 0.001
IQR outlier bounds	[−5,156.5, 15,207.5] → 0 outliers

Honest read of the shape: the near-zero skew (−0.05) makes the distribution look symmetric, but the negative kurtosis (−1.20) and the Shapiro-Wilk test ($p < 0.001$) both say this is *not* a normal bell curve. It's flatter and closer to uniform, amounts are spread fairly evenly from about 100 to 10,000 rather than clustering around the mean. The box plot shows a wide IQR (2,480 to 7,571) with zero statistical outliers, consistent with a roughly uniform spread rather than a peaked distribution. I flagged this rather than describing it as "normal" just because the question says "bell curve."

Q4. Monthly Cohort View of Active Users (DEBIT Transactions)

Cohort \ Active Month	2020-01	2020-02	2020-03	2020-04	2020-05	2020-06	2020-07
2020-01	8	5	8	7	7	7	8
2020-02	0	1	0	1	1	0	1
2020-03	0	0	1	1	1	1	1
2020-04	0	0	0	0	0	0	0
2020-05	0	0	0	0	0	0	0
2020-06	0	0	0	0	0	0	0
2020-07	0	0	0	0	0	0	0

Same table as retention percentage (of cohort's original size)

Cohort \ Active Month	2020-01	2020-02	2020-03	2020-04	2020-05	2020-06	2020-07
2020-01 (n=8)	100%	62.5%	100%	87.5%	87.5%	87.5%	100%
2020-02 (n=1)	—	100%	0%	100%	100%	0%	100%
2020-03 (n=1)	—	—	100%	100%	100%	100%	100%

Definitions used: a user's cohort month is the calendar month of their first DEBIT transaction. "Active" in a given month means they placed at least one DEBIT transaction that month (per the question's own parenthetical: "Users doing DEBIT txns"). All 10 users in the dataset had their first debit in Jan, Feb, or March 2020, none first appear from April onward, which is why rows 4 to 7 are entirely zero rather than missing data. Retention for the January cohort stays high (87.5 to 100%) through July, meaning this is a small, highly-retained user base rather than typical consumer-app churn.

Q5. Top 10 Percentile Users by Net Amount (DEBIT − CREDIT)

```
WITH user_net AS (
  SELECT user_id,
         SUM(CASE WHEN txn_type='DEBIT' THEN txn_amt ELSE 0 END) -
         SUM(CASE WHEN txn_type='CREDIT' THEN txn_amt ELSE 0 END) AS net_amount
```

```
  FROM transactions
  GROUP BY user_id
),
ranked AS (
  SELECT user_id, net_amount,
         PERCENT_RANK() OVER (ORDER BY net_amount DESC) AS pct_rank
  FROM user_net
)
SELECT user_id, net_amount
FROM ranked
WHERE pct_rank <= 0.10
ORDER BY net_amount DESC;
```

Answer: User 8, Net Amount = −35,804

Rank	User ID	Net Amount (DEBIT−CREDIT)
1	**8**	**−35,804**
2	1	−37,313
3	3	−39,640
4	10	−58,427
5	4	−88,041
6	7	−113,196
7	6	−117,389
8	9	−127,790
9	2	−132,453
10	5	−147,655

Caveat worth stating directly: the dataset only has 10 unique users. "Top 10 percentile" of 10 users is mathematically just the single highest-ranked user (User 8), not a meaningfully-sized cohort. I ran the query correctly using `PERCENT_RANK()` so the logic generalizes to a larger user base, but on this specific dataset it reduces to n=1. I'm flagging that rather than presenting a 1-row table as if it came from a large enough sample to trust.

Also worth noting: every user's net amount (DEBIT−CREDIT) is negative, meaning every user in this dataset was a net receiver of funds (CREDIT total > DEBIT total) across Jan–Jul 2020. If "top" was intended to mean "highest net spender" (largest positive DEBIT−CREDIT), no user qualifies; if it means "closest to zero / least net-negative," User 8 is the answer above.